

Too Hot to Work? Heat Waves and Construction Laborers in the U.S.

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October 2026

Abstract

This study investigates the impact of heat waves on the working hours of construction laborers in U.S. metropolitan statistical areas (MSAs). Utilizing daily temperature data and working hours data, the research explores how extreme heat affects labor output in short-term heat shocks, contrasting with prior studies that often focus on long-term climate adaptation. Using a two-way fixed effects model, this paper investigates the impact of increasing heat waves on construction laborers, accounting for both individual and temporal heterogeneity. The results suggest that in samples restricted to weeks with low precipitation, an additional day per week with temperatures exceeding 90°F reduces working hours by about 0.22 hours, and the occurrence of heat waves at least once a week decreases working hours by about 0.75 hours. Despite the statistical significance of some results, implications must be drawn carefully as the magnitudes are not large.

Keywords: heat waves, working hours, construction laborers, temperature shock

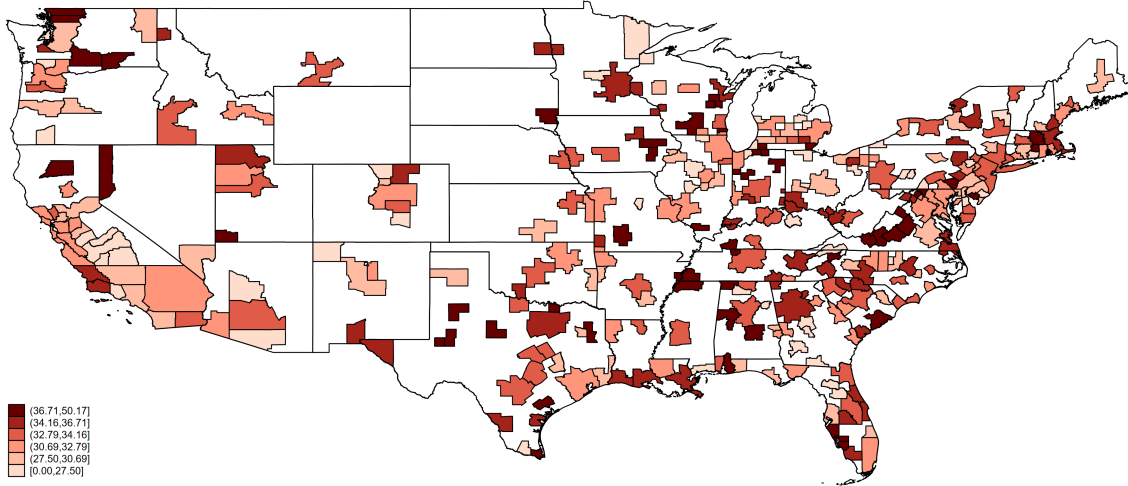
JEL Classification: J20, Q54

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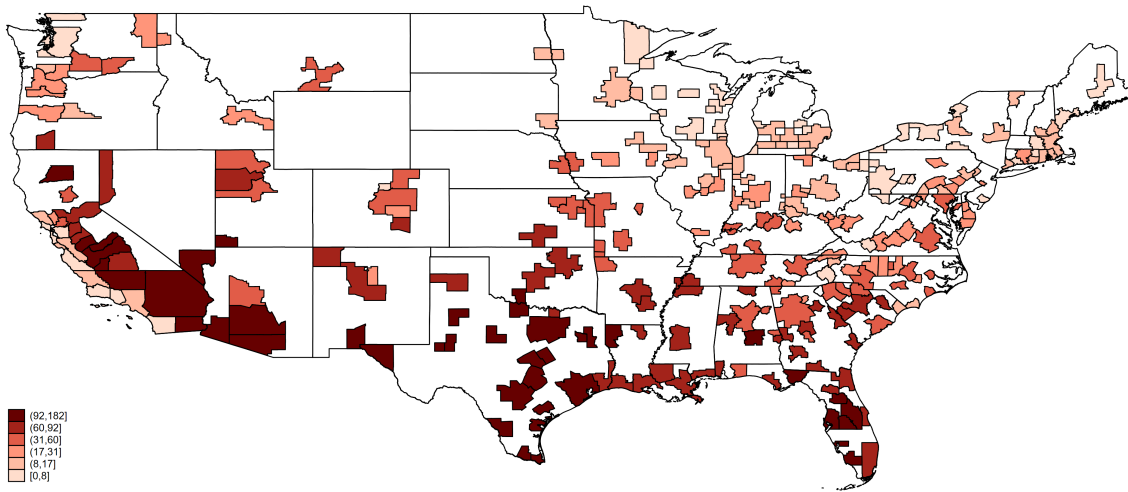
1. Introduction

Global warming has been a focal point of significant concern for several decades. While this phenomenon represents a long-term climatic shift, it also contributes to more frequent and intense short-term weather events, particularly heat waves. These periods of extreme heat have recently emerged as a critical issue with immediate and localized impacts. The primary distinction between these two phenomena lies in their definitions and effects: global warming signifies a sustained increase in Earth's average surface temperature, exerting a gradual influence across the globe. In contrast, heat waves are characterized by transient episodes of abnormally hot weather, often accompanied by elevated humidity levels, delivering acute, geographically concentrated impacts. Given their sudden onset and potential for harm, heat waves are more akin to extreme weather events or natural disasters. Additionally, once heat stress surpasses a specific threshold, the physiological mechanisms that regulate internal body temperature fail to maintain it within the requisite range for normal functioning ([ILO, 2019](#)). It is therefore essential to distinguish between global warming and heat waves in scholarly analyses, especially considering the diverse implications they may have on human economic behaviors. This paper is driven by an interest in exploring how changes in heat wave intensity over time might alter individual behaviors, particularly in scenarios where the workplace environment becomes unbearably hot. Using the daily temperature and the working hours of construction workers in over 250 MSAs, I investigate the impact of heat waves on the labor supply of construction laborers in the U.S.

Contrary to the common assumption that heat waves primarily affect Eastern Europe and Southeast Asia, recent empirical evidence has also confirmed their significant presence in the United States. Importantly, these heat waves display considerable regional variation within the U.S. As illustrated in Figure 1, there is a discernible disparity not only in the working hours but also in the heat wave frequency. The first map describes the average working hours of all workers in U.S. metropolitan areas, while the second map shows the



(a) Average weekly working hours



(b) Average annual number of days with maximum temperature above 90°F

Figure 1. Working Hours and Heat Wave Frequency across MSAs

regional distribution of the yearly average of the number of days when the temperature exceeds 90°F. One noticeable point is that the MSAs with higher working hours are mainly concentrated from the Midwest to the East, whereas the areas with more frequent heat waves are concentrated in the southwestern part of the country.

The existing body of literature predominantly examines the impact of heat waves on economic activity, mainly focusing on output loss, particularly within the agricultural sector. Notably, [Mendelsohn et al. \(1994\)](#) employed a Ricardian approach to demonstrate how adaptive agricultural practices might mitigate the adverse economic consequences

of climate change on U.S. agriculture (see also [Mendelsohn et al., 2000](#); [Roberts and Schlenker, 2011](#)). Some studies, including [Sudarshan and Tewari \(2014\)](#), have broadened the focus to include non-agricultural impacts of heat waves. For instance, [Deschênes and Greenstone \(2011\)](#) forecast an increase in U.S. mortality rates and residential energy consumption by the end of the century due to climate change. Despite their contributions to understanding the impact of gradual temperature changes, these studies overlooked the economic implications of abrupt temperature shocks.

Recent research has begun to explore the economic effects of excessive heat. [Cachon et al. \(2012\)](#) analyzed how heat stress affects productivity in manufacturing, a sector characterized by intensive labor demands and vulnerability to heat exposure ([ILO, 2019](#)). Although they shed light on the consequences of extreme heat on production, they focused on the change in total output rather than on labor productivity. [Graff Zivin and Neidell \(2014\)](#), using data from the American Time Use Survey (ATUS), demonstrated that workers highly exposed to weather conditions experienced reductions in working hours of up to 59 minutes when daily maximum temperatures exceeded 85°F. [Park \(2016\)](#) examined the long-run adaptive responses of labor production to heat shocks, finding that an increase in the frequency of heat waves leads to a decrease in labor output measured by payroll per capita. However, the paper oversimplifies the damage of heat waves to non-labor inputs, which may result in overestimating the impact of extreme heat on labor productivity. In contrast, this research concentrates on the direct impact of heat waves on workers by narrowing down the scope from all sectors to construction sites, where not only are the workers highly exposed to extreme heat, but the occupation is also labor-intensive.

This study aims to assess the immediate effects of heat waves on labor supply, with a particular focus on construction laborers in U.S. metropolitan areas. It provides an analysis of the short-term responses of workers to extreme heat conditions. Unlike prior studies that concentrate on long-term adaptations, this paper quantifies the direct impacts experienced by workers during heat waves, offering insights into their sensitivity to

extreme temperatures and providing further implications for the necessity of adaptation for workers who are relatively more exposed and vulnerable to heat shocks. Additionally, by studying the United States, a developed country with abundant resources for mitigating heat stress such as air conditioning, the findings may highlight greater sensitivities to heat waves compared to similar analyses in labor-intensive industries within developing countries. By focusing specifically on construction workers, who are acutely susceptible to heat waves, this research also aims to delineate the direct effects of heat stress on labor productivity.

It is critical to define how to measure the heat wave. Frequency, defined as the number of days in which the temperature exceeds a specific threshold, may be more adequate for assessing the impact of heat waves on labor supply compared to intensity, which measures how hot it is during the heat wave. This is because maintaining a body temperature around 98°F is essential for proper human functioning, and bodily functions decline significantly when temperatures exceed a certain point. Furthermore, multiple issues must be discussed regarding the definition of a heat wave. The first point is whether to use the maximum or minimum temperature when measuring heat waves. For this research, I use the maximum daily temperature instead of the minimum temperature. Although some argue that extreme heat at bedtime is more life-threatening due to the potential for severe sleep deprivation caused by nighttime heat waves, I focus on the maximum temperature. This is because the maximum temperature, typically observed around noon, is more likely to directly affect the workplace conditions of construction workers and less likely to be affected by adaptations such as air conditioning.

The second point concerns the establishment of the threshold temperature, which is crucial for measuring the frequency of heat waves. In this paper, I propose using a global threshold across all regions instead of a local threshold that accounts for regional climate differences. While focusing on a local threshold might be more appropriate since extreme heat is a type of extreme weather, measured by the thickness of the tails of the weather

distribution, it is implausible to assume that individuals in the coldest regions experience similar degrees of extreme heat as those in the hottest regions. Third, one must be cautious about setting the degree of the threshold temperature, as it significantly affects the main results. This paper provides results derived from multiple threshold temperatures to prevent the research results from being contingent on the degree of the threshold.

This study unveils the measurable effects of heat waves on the working hours of construction laborers in U.S. metropolitan areas. The findings indicate that heat waves do impact labor supply, but the effects are modest and context-dependent. Specifically, I find that an additional day per week with temperatures exceeding 90°F reduces working hours by about 0.22 hours. Moreover, experiencing heat waves at least once a week decreases working hours by about 0.75 hours. Importantly, these effects are statistically significant only under conditions of low precipitation, suggesting a complex interplay between high temperature and other weather factors, including rainfall. The relatively small magnitude of these effects may indicate the presence of adaptation mechanisms in the construction industry. However, as climate change intensifies and heat waves become more frequent, even small reductions in working hours could accumulate substantial economic impacts over time. These findings emphasize the need for adaptive strategies to protect vulnerable workers and maintain the labor supply in changing climate conditions.

The remainder of this paper is organized as follows. Section 2 describes the data. Section 3 outlines the empirical strategy. Section 4 presents the findings, and Section 5 concludes.

2. Data

2.1 Daily Temperature

The data employed in this study to measure heat waves are sourced from the Global Historical Climatology Network Daily (GHCN-Daily) dataset, Version 3, provided by the National Centers for Environmental Information (NCEI). This dataset combines daily

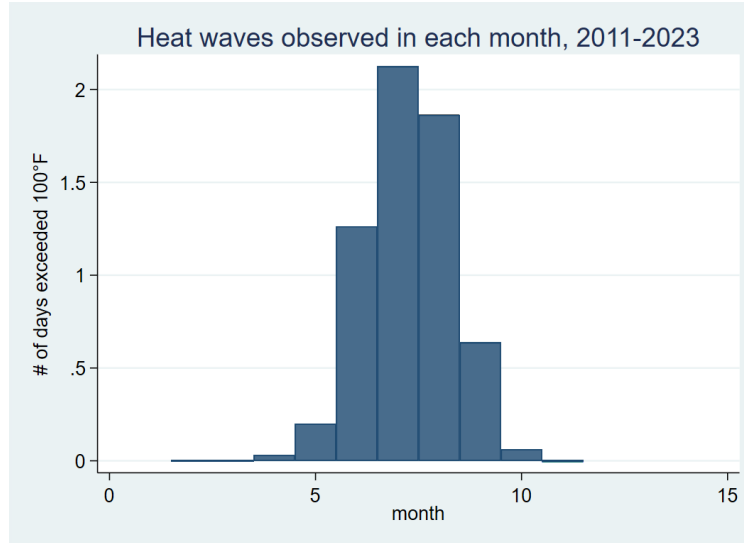


Figure 2. Heat Waves Observed in Each Month, 2011–2023

weather measurements from approximately 30 distinct sources, encompassing variables such as precipitation, daily high and low temperatures, temperature at the time of observation, snowfall, and snow depth. These data are collected from over 90,000 terrestrial stations worldwide, with all temperature readings measured in Celsius. For this analysis, I utilize maximum temperature data from stations located within U.S. territory, so each observation of the daily maximum temperature of a metropolitan area is the average of the daily maximum temperatures recorded by the stations within the area. Although the raw data contain weather information for all 12 months from 2011 to 2023, I employ the data collected in May, June, July, August, and September. This temporal focus on five months of each year is designed to capture the peak months when heat waves are most likely to occur. Figure 2 shows that heat waves are mainly concentrated in the target months. For example, the average reference week in July included approximately two days on which temperatures exceeded 100°F.

Using this comprehensive dataset, I calculate the number of days on which the maximum temperature exceeded preset thresholds, thereby establishing a monthly frequency of heat waves for each MSA. Specifically, I employ three different thresholds: 90°F, 95°F, and 100°F.

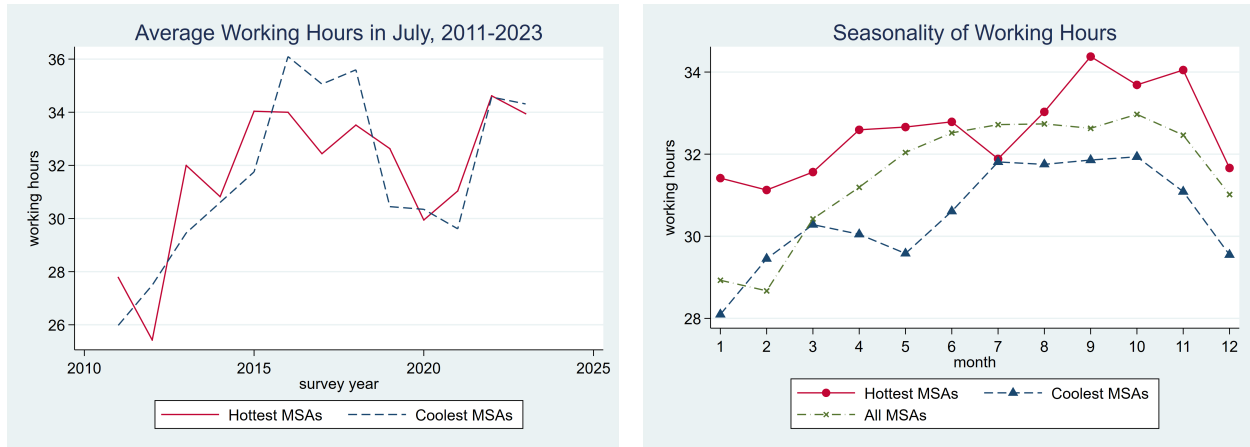
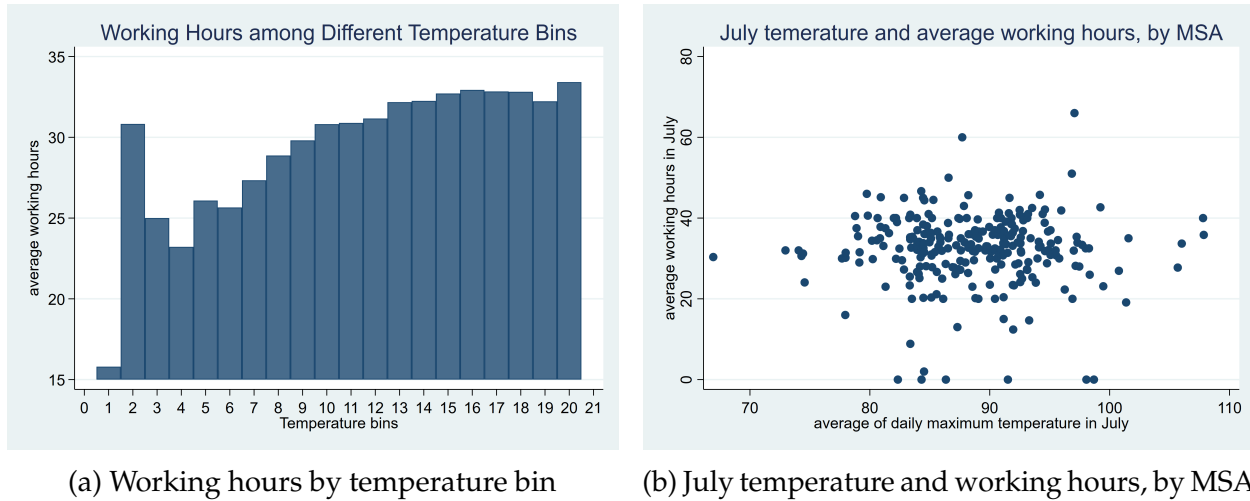


Figure 3. Working Hours in the Hottest and Coolest MSAs

Notes: The hottest (coolest) group consists of MSAs in the top (bottom) decile of average July maximum temperature. The left panel plots average July working hours of construction laborers by survey year. The right panel plots average working hours by calendar month.

Figure 3 illustrates the characteristics of the data by concentrating on two decile groups of MSAs. The graph on the left shows the trend of average working hours in July for construction laborers living in the hottest and coolest MSAs in the United States. Specifically, the hottest group consists of MSAs where the average maximum temperature in July is in the top 10%, while the coolest group includes MSAs in the bottom 10%. Although a common trend in working hours is observed in both groups, the distinction between them remains unclear. The graph on the right illustrates the seasonality of construction laborers' working hours. Both the overall average seasonality and the seasonality in the coolest MSAs suggest that summer and fall are the most favorable times for work. Notably, in the hottest areas, the monthly trend shows that in July, when heat waves are most prevalent, the average working hours of construction workers are lower than in the adjacent months. Nevertheless, conclusions should be drawn with extra care.

Figure 4a illustrates the relationship between the average maximum temperature and average working hours of construction laborers. The x-axis represents temperature bins, each spanning 5°F, with Bin 1 representing temperatures below 5°F and Bin 20 representing temperatures above 100°F. The y-axis shows average weekly working hours. The figure shows a clear upward trend in working hours as temperatures increase from the



(a) Working hours by temperature bin (b) July temperature and working hours, by MSA

Figure 4. Temperature and Working Hours of Construction Laborers

coldest bins to moderately warm bins. This suggests that construction activity generally increases in warmer weather, likely due to more favorable working conditions. However, as temperatures increase beyond 80°F, the graph shows a slight plateau and potential decline in working hours. This could indicate the onset of heat stress effects on labor supply, where extremely high temperatures begin to negatively impact workers' ability to maintain long working hours.

Figure 4b is a scatterplot of MSAs drawn with the combination of average July maximum temperature and average working hours of construction laborers in each MSA in July. The points are concentrated around a horizontal line, indicating that average working hours are around 35. Meanwhile, the average maximum temperature in July is widely dispersed across MSAs, making it difficult to detect any relationship between the average maximum temperature and average working hours.¹

¹Average working hours are zero in some MSAs regardless of the average maximum temperature because this paper treats observations recorded as "Not In Universe" as unemployed workers, assuming their actual working hours are zero. On one hand, it is plausible to treat those observations as unemployed in the sense that the question asking about actual working hours last week may not apply to unemployed workers, meaning that they worked zero hours. On the other hand, such a transformation could also be one limitation of this research.

2.2 Hours Worked

The hours worked data utilized in this study are sourced from the IPUMS Current Population Survey (CPS), which standardizes microdata from the monthly U.S. labor force survey. CPS respondents are asked to recall information about events that occurred during the previous week (the “reference period”), the week before the calendar week that includes the 19th. Therefore, when integrating the two datasets, I specifically utilize daily temperature data from the week that contains the 12th day of each month, which coincides with the CPS reference period.

The CPS dataset is rich in demographic details but is particularly valuable for this study due to its inclusion of actual working hours, which serve as the dependent variable in my estimation. For this analysis, I use data on workers residing in 267 MSAs, allowing for a focused examination of labor dynamics within each region. Specifically, the dataset includes 674 occupations classified under 223 industries.

A notable advantage of using CPS data is its longitudinal tracking of individuals over the same season across two consecutive years. This tracking is facilitated by the CPS’s unique survey design, where data are collected from individuals for four consecutive months, followed by an eight-month break, and then resumed for another four months. Consequently, the dataset naturally captures information from a sample of individuals across two consecutive summers.

Table 1 presents the summary statistics for selected variables within the dataset. On average, construction laborers work 32.62 hours per week in the summer, experiencing about 2.4 days per week on which the maximum temperature exceeds 90°F. Alternatively, the probability that an average construction worker in a metropolitan area will experience a heat wave is approximately 0.55. Notably, there is significant heterogeneity in working hours among workers. Some respondents reported working more than 40 hours a week, and some even reported more than 168 hours, which exceeds the number of hours theoretically available. To mitigate distortion caused by outliers, potentially arising from

Table 1. Summary Statistics

	Mean	SD	Min	Max
Actual working hours	32.62	16.15	0	70
<i>Frequency of heat waves (days per reference week)</i>				
Above 90°F	2.402	2.722	0	7
Above 95°F	0.958	1.956	0	7
Above 100°F	0.382	1.298	0	7
<i>Occurrence of heat waves (indicator)</i>				
Above 90°F	0.553	0.497	0	1
Above 95°F	0.264	0.441	0	1
Above 100°F	0.112	0.315	0	1

Notes: The sample consists of construction laborers observed in the CPS reference week (the week containing the 12th day of each month) from May to September of each year, 2011–2023. Observations reporting more than 70 weekly working hours (the 99th percentile) are excluded.

measurement errors, this study introduces an upper bound for working hours based on their distribution. Specifically, the upper bound is set at the 99th percentile of working hours among construction laborers, identified as 70 hours per week. The 223 cases in which working hours exceeded 70 hours per week are truncated, which accounts for 0.76% of construction laborers.²

3. Empirical Strategy

An ideal experiment to assess the impact of heat waves on workers would involve establishing two identical areas, subjecting only one to excessive heat, and observing differences in working hours between the two. However, since replicating such conditions in reality is not feasible, a quasi-experimental alternative is employed. This paper employs the two-way fixed effects (TWFE) model for identification, drawing inspiration from [Deschênes and Greenstone \(2007\)](#), who used the TWFE model to address economic questions related

²The possible outliers are excluded from the analysis as the true working hours of those construction laborers are unknown. Nevertheless, the result could be largely affected by the truncation. This paper provides results with alternative upper bounds, which are the top 5%, 0.5%, and 0.1% of the working hours of construction laborers. Additionally, the appendix also provides estimation results when censoring working hours at 70 hours per week. This approach accounts for the possibility that the true working hours of the truncated observations may exceed the mean so that the gap between the true working hours and the reported values may be small.

to global warming. The estimating equation is

$$Hours_{it} = c + \beta HeatWave_{it} + \alpha_i + \delta_t + \epsilon_{it}, \quad (1)$$

where $Hours_{it}$ represents the actual weekly working hours of construction laborer i in year-month t . The variable $HeatWave_{it}$ denotes the frequency of heat waves, measured by the number of days in reference week t that exceeded the threshold temperature. I also introduce an alternative independent variable in which $HeatWave_{it}$ is binary, so that $HeatWave_{it} = 1$ implies that a heat wave was observed within the reference week of each year-month and $HeatWave_{it} = 0$ otherwise. Given the severe physiological effects on the human body at temperatures beyond its maximum tolerance threshold, employing a binary variable simplifies the empirical model. However, this approach has the drawback of overlooking the relationship between the intensity of heat waves and working hours, so one should be cautious when comparing the results from the two kinds of explanatory variables. In both the continuous and binary models, the parameter β captures the causal impact of heat waves on the labor supply of construction workers.

The model also includes δ_t to account for time fixed effects, specifically year-month fixed effects. This parameter is expected to capture the impact of macroeconomic conditions and the seasonality of weather on working hours. The parameter α_i denotes the individual fixed effect, encompassing factors such as skills, effort, or education that differ among individuals. The error term ϵ_{it} captures the unobserved factors affecting individuals' labor supply decisions. Since individual fixed effects and time fixed effects account for individual heterogeneity and individual-invariant heterogeneity, respectively, the error term is expected to include only the unobservables that vary among individuals in the short run (e.g., fluctuations in daily physical condition). Therefore, the error term is expected to be uncorrelated with heat waves. Standard errors are clustered at the MSA level.³

³This paper clusters standard errors by MSA since each MSA is assumed to be equally treated by the

Table 2. Impact of Heat Waves on Working Hours of Construction Laborers

Threshold of heat wave	Working hours of construction laborers					
	90°F (1)	95°F (2)	100°F (3)	90°F (4)	95°F (5)	100°F (6)
Frequency of heat waves	−0.069 (0.057)	0.018 (0.060)	−0.052 (0.100)			
Occurrence of heat waves				0.014 (0.248)	0.100 (0.301)	−0.215 (0.418)
Observations	22,100	22,100	22,100	22,100	22,100	22,100

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table presents two-way fixed effects estimates for construction laborers in 257 metropolitan areas from May 2011 to September 2023, using different thresholds to define a heat wave. Frequency is the number of days in the reference week on which the maximum daily temperature exceeds each threshold. Occurrence is an indicator equal to one if at least one day in the reference week has a maximum temperature exceeding the threshold. The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

4. Results

Table 2 presents the results of the empirical analysis using two-way fixed effects models with different explanatory variables. Columns (1) to (3) use the frequency of heat waves as the independent variable, while columns (4) to (6) focus on the occurrence of heat waves instead.

The implications drawn from Table 2 should be interpreted cautiously, as the estimates from all models are statistically insignificant. Nevertheless, a few possible explanations could support these findings. First, it might indicate that construction laborers in the United States are leveraging various adaptations, such as using shelters with air conditioning or adopting flexible work schedules that avoid the hottest times of the day. Second, it is possible that the disutility of construction laborers is not reflected in actual working hours, depending on how labor contracts are made. For instance, if labor hours are predetermined by contracts between the laborers and the construction company prior to construction, working hours might appear unaffected by heat waves. Third, it could be a result of

same amount of heat wave. One could also consider clustering standard errors by time, in the sense that all individuals undergo the same seasonal change of the weather, but it is less persuasive considering that the geographical variance of temperature is non-negligible.

Table 3. Impact of Heat Waves on Working Hours, Controlling for Precipitation

Threshold of heat wave	Working hours of construction laborers					
	90°F (1)	95°F (2)	100°F (3)	90°F (4)	95°F (5)	100°F (6)
Frequency of heat waves	−0.090 (0.057)	0.004 (0.062)	−0.065 (0.100)			
Occurrence of heat waves				−0.012 (0.248)	0.069 (0.306)	−0.250 (0.420)
Precipitation	−0.160 (0.135)	−0.125 (0.133)	−0.133 (0.131)	−0.126 (0.132)	−0.124 (0.132)	−0.130 (0.130)
Observations	22,100	22,100	22,100	22,100	22,100	22,100

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table replicates Table 2, additionally controlling for precipitation, defined as the total amount of precipitation (inches) in the reference week of each month. The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

misspecification, particularly omitted variables. As construction laborers work outdoors, they are exposed to other weather events like rainfall, which increases humidity and consequently creates an unpleasant environment for the laborers. To reinforce the analysis, I add the total amount of precipitation in each reference week as a control variable.⁴ Precipitation could affect the estimation results, as extreme heat combined with high humidity harms the human body (ILO, 2019). On the other hand, rainfall can also mitigate a heat wave by cooling the temperature, providing additional grounds for including precipitation in the empirical model.

Contrary to the perception that precipitation would severely discourage construction laborers, Table 3 suggests that the impact is ambiguous. As in the case of heat waves, this could be the result of adaptation, such as introducing a flexible construction schedule to conduct construction only on clear days, which leaves weekly working hours unaffected.

An alternative approach to control for the effect of precipitation on working hours is to limit the sample to instances with minimal or no precipitation. Table 4 presents

⁴However, precipitation should be carefully included in the empirical model due to the potential correlation between temperature and precipitation. Generally, higher surface temperatures lead to increased evaporation, which in turn results in greater precipitation.

the estimation results using the same empirical model as in Table 2, but the sample is restricted to cases where the amount of precipitation falls below the median of the precipitation distribution.⁵ Interestingly, the estimates in columns (1) and (4) imply that heat waves negatively impact the working hours of construction laborers, as the estimates are statistically significant at the 5% level. Specifically, an average construction laborer works about 0.22 hours less for each additional day on which the maximum temperature exceeds 90°F. Furthermore, experiencing heat waves at least once a week decreases working hours by about 0.75 hours per week. The lower endpoint of the 95% confidence interval for column (4) is -1.508 , so the data rule out weekly reductions larger than about 1.5 hours at conventional confidence levels. Yet, the small sizes of the estimated coefficients in columns (1) and (4) indicate that construction laborers have little incentive to decrease their working hours under heat waves. The result might imply that heat waves do not have a lethal impact on construction laborers, but instead impose a tolerable disutility. However, one should keep in mind that heat waves may be an endurable pain to construction laborers because of adaptation mechanisms, not because heat waves deal little damage to human bodies.

Table 5 reports the effects of heat waves on the construction industry and other occupations. Column (2) presents estimates of how the working hours of workers in the entire construction industry are affected by the frequency of heat waves. The statistical insignificance of the estimate in column (2) could arise because the construction industry includes occupations other than construction laborers, such as construction managers or tradespeople, who are less likely to work outside for the entire working day. Columns (3) and (4) provide estimates from the same model but use observations on farmers and accountants, respectively. Farmers, like construction laborers, are likely to work outside, whereas accountants typically work indoors. However, the data do not specify how long

⁵It may sound arbitrary that the threshold is set at the 50th percentile of precipitation, or precisely 0.417 inches. Considering that the mean of precipitation is 0.808 inches, however, it could be a reasonable upper bound to define “low precipitation.” As a comparison, the appendix reports alternative results for different thresholds of low precipitation.

Table 4. Impact of Heat Waves on Working Hours under Low Precipitation

Threshold of heat wave	Working hours of construction laborers					
	90°F (1)	95°F (2)	100°F (3)	90°F (4)	95°F (5)	100°F (6)
Frequency of heat waves	−0.221** (0.091)	−0.059 (0.078)	−0.081 (0.119)			
Occurrence of heat waves				−0.754** (0.383)	−0.193 (0.407)	−0.301 (0.417)
Observations	9,387	9,387	9,387	9,387	9,387	9,387

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table replicates Table 2, restricting the sample to observations with precipitation below the median (0.417 inches). The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

Table 5. Impact of Heat Waves on Working Hours across Occupations and Industries

	Working hours			
	Construction laborers (1)	Construction industry (2)	Farmers and agric. managers (3)	Accountants and auditors (4)
Frequency of heat waves	−0.221** (0.091)	0.034 (0.049)	0.147 (0.232)	0.011 (0.099)
Observations	9,387	60,266	2,786	9,903

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table presents two-way fixed effects estimates for workers in three occupations and in one broader category (the construction industry) from May 2011 to September 2023. As in Table 4, the sample is restricted to observations with precipitation below the median. Frequency is the number of days in the reference week on which the maximum daily temperature exceeds 90°F. The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

farmers or accountants work outside or inside during their working hours, which may explain why the estimates are ambiguous and statistically insignificant.

5. Conclusion

This paper examines the impact of heat waves on the working hours of construction laborers in U.S. metropolitan areas using a two-way fixed effects model. The findings reveal that an additional day per week with temperatures exceeding 90°F reduces working hours by about 0.22 hours, and experiencing heat waves at least once a week decreases

working hours by about 0.75 hours. However, these findings are statistically significant only when the analysis is restricted to periods of low precipitation, highlighting the complex relationship between heat waves, precipitation, and labor supply.

Although the magnitudes are small, the study underscores the specific vulnerability of construction workers to extreme heat. The need for targeted interventions and adaptive strategies to mitigate these adverse effects is evident as global temperatures continue to rise, which also implies the intensification of heat waves.

Despite the findings, this paper has limitations that leave room for further development. Particularly with the binary independent variable of heat waves, staggered intervention issues are highly likely to be hidden under this empirical model. The impact of intense heat shocks may vary across individuals and time periods, potentially challenging the parallel trends assumption underlying the difference-in-differences approach. Future research should explore methods to address this potential heterogeneity in treatment effects. Taking various types of adaptation into consideration is another task to be addressed in future development of this research. From air conditioning systems and weather forecasts to voluntary resignation or a change in career, it is necessary to address as many kinds of adaptation as possible in the model, as major yet unknown adaptation mechanisms could cause omitted variable bias.

As global temperatures continue to rise, addressing the challenges posed by heat waves will be essential for ensuring the health, safety, and productivity of the workforce. Currently, there is a significant gap in the frequency of heat waves between the 90°F and 100°F thresholds, and the number of areas that exhibit extreme levels of heat waves is still limited. However, the acceleration of global warming will also increase the frequency of extremely hot weather. Future research should aim to explore not only the long-term adaptive measures that have been extensively studied but also the short-term adaptations that protect workers who are directly exposed to high temperatures from heat-related illnesses.

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A. Additional Results

A.1 Alternative Truncation Thresholds

Panels A and B of Table [A1](#) report the regression results with different thresholds for identifying outliers with extreme values of working hours. Specifically, Panel A uses the same model as Table [2](#), but the truncation threshold of working hours is the top 5%, whereas Panel B uses the top 0.5% as the threshold. The results are statistically insignificant and the magnitudes of the estimates in each column remain small, as in Table [2](#).

A.2 Censoring versus Truncating

If one believes that the true working hours of the outlying construction laborers are high, censoring those outliers could be a better option than truncating. Panel C of Table [A1](#) shows the regression results when the working hours of the outliers are censored at 70 hours. Despite the change in the magnitudes of the coefficients, censoring the data has only a minor effect on the statistical significance of the results compared to Table [2](#).

A.3 Alternative Definitions of Low Precipitation

The results of Table [4](#) raise the question of whether the estimates differ substantially if low precipitation is defined otherwise. Table [A2](#) reports the results using various thresholds for determining low precipitation. For simplicity, the table only reports results where the heat wave threshold is 90°F. The smaller the threshold, the more restricted the sample. Limiting the sample to weeks with low precipitation strengthens the statistical significance of the estimates, but an excessive restriction weakens it instead. One implication is that heat stress is magnified under lower levels of humidity, which supports the idea that precipitation cools down heat waves and mitigates their effect. However, the impact of heat waves becomes ambiguous if the weather is too dry. This might be due to the much smaller sample size when a stricter threshold is applied.

Table A1. Alternative Treatments of Outlying Working Hours

Threshold of heat wave	Working hours of construction laborers					
	90°F (1)	95°F (2)	100°F (3)	90°F (4)	95°F (5)	100°F (6)
<i>Panel A: Truncation at the top 5%</i>						
Frequency of heat waves	−0.086 (0.054)	−0.027 (0.059)	−0.068 (0.089)			
Occurrence of heat waves				−0.008 (0.247)	−0.031 (0.277)	−0.304 (0.418)
Observations	21,046	21,046	21,046	21,046	21,046	21,046
<i>Panel B: Truncation at the top 0.5%</i>						
Frequency of heat waves	−0.064 (0.057)	0.019 (0.060)	−0.052 (0.098)			
Occurrence of heat waves				0.030 (0.249)	0.110 (0.300)	−0.204 (0.416)
Observations	22,167	22,167	22,167	22,167	22,167	22,167
<i>Panel C: Censoring at 70 hours (top 1%)</i>						
Frequency of heat waves	−0.055 (0.055)	0.032 (0.061)	−0.047 (0.094)			
Occurrence of heat waves				0.040 (0.250)	0.106 (0.307)	−0.224 (0.398)
Observations	22,315	22,315	22,315	22,315	22,315	22,315

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table replicates Table 2 under alternative treatments of observations with extreme working hours. Panels A and B drop observations above the 95th and 99.5th percentiles of working hours, respectively. Panel C retains all observations and censors working hours at 70 hours (the 99th percentile). The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

A.4 MSA Fixed Effects versus Individual Fixed Effects

The main specification includes individual fixed effects in addition to time fixed effects, based on the assumption that heterogeneity is expected at the individual level. Nevertheless, one might suspect that heterogeneity among metropolitan areas is significant while heterogeneity among individuals within a given MSA is not. As a robustness check, Table A3 reports results that include MSA fixed effects instead of individual fixed effects. Panel A uses the full sample and is paired with Table 2, while Panel B restricts the sample to observations with low precipitation and is paired with Table 4. The comparison shows that the magnitudes of the estimated coefficients remain small. Moreover, the results in

Table A2. Alternative Thresholds for Low Precipitation

Low precipitation cutoff (percentile of precipitation)	Working hours of construction laborers					
	30th (1)	50th (2)	70th (3)	30th (4)	50th (5)	70th (6)
Frequency of heat waves	−0.176 (0.124)	−0.221** (0.091)	−0.197*** (0.073)			
Occurrence of heat waves				−0.091 (0.467)	−0.754** (0.383)	−0.552* (0.308)
Observations	5,695	9,387	13,918	5,695	9,387	13,918

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table replicates columns (1) and (4) of Table 4, defining low precipitation as precipitation below the 30th, 50th, or 70th percentile. The heat wave threshold is 90°F in all columns. The table reports estimated coefficients with standard errors in parentheses. All models include individual and year-month fixed effects. Standard errors are clustered at the MSA level.

Table A3. Specifications with MSA Fixed Effects

Threshold of heat wave	Working hours of construction laborers					
	90°F (1)	95°F (2)	100°F (3)	90°F (4)	95°F (5)	100°F (6)
<i>Panel A: Full sample</i>						
Frequency of heat waves	−0.026 (0.065)	−0.059 (0.079)	−0.043 (0.171)			
Occurrence of heat waves				−0.088 (0.305)	−0.169 (0.275)	0.267 (0.577)
Observations	28,749	28,749	28,749	28,749	28,749	28,749
<i>Panel B: Low precipitation sample</i>						
Frequency of heat waves	−0.120 (0.088)	−0.081 (0.098)	−0.015 (0.169)			
Occurrence of heat waves				−0.070 (0.538)	−0.480 (0.352)	0.384 (0.561)
Observations	15,102	15,102	15,102	15,102	15,102	15,102

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Notes: This table replicates Table 2 (Panel A) and Table 4 (Panel B), replacing individual fixed effects with MSA fixed effects. The table reports estimated coefficients with standard errors in parentheses. All models include MSA and year-month fixed effects. Standard errors are clustered at the MSA level.

Table 4 lose their statistical significance when MSA fixed effects replace individual fixed effects. This may be evidence that individual heterogeneity is more pronounced than heterogeneity at the MSA level.